Health & Place 99 (2026) 103660



Contents lists available at ScienceDirect

Health and Place

journal homepage: www.elsevier.com/locate/healthplace





Green space, life-course socioeconomic disparities and cognitive function in later life: A longitudinal analysis

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ARTICLE INFO

Keywords:
Green space
Life-course perspective
Socioeconomic disparities
Cognitive function
Older adults

ABSTRACT

There is evidence that green space may reduce socioeconomic disparities in health ('equigenesis'). However, it is still unclear whether such a finding is also valid for cognitive function in older adults. Therefore, using data from the Northern Ireland Cohort for the Longitudinal Study of Ageing, this study examines whether exposure to green spaces reduces the effect of socioeconomic disparities in cognitive function in later life. It uses the Normalised Difference Vegetation Index (NDVI) and accessibility of green space (distance to the nearest parks/woodland) as estimates of green space exposure. Cognitive function in later life was measured using the Mini-Mental State Examination (MMSE) scores. Life-course disparities were assessed in early life (self-reported childhood socio-economic status and childhood unemployment of parents), and later life (education, Area-level Multiple Deprivation Measure and capital value of the housing), respectively. Multilevel linear regressions were used to model the associations among green space, life-course socioeconomic disparities and cognitive function. Results show that MMSE scores were positively associated with NDVI (Coef. = 0.51; SE = 0.13), but negatively associated with distance to the nearest parks (Coef. = −0.09, SE = 0.02). Also, participants with higher educational attainment, living in a property with a higher capital value or in less deprived neighbourhoods had higher MMSE scores. The association effects of green space exposure were stronger in older adults with lower socioeconomic status in early and later life. Our findings suggest that green space may reduce the life-course socioeconomic inequalities in cognitive function in later life.

1. Introduction

Based on a report from the World Health Organisation, from 2015 to 2050, the proportion of the global ageing population is projected to grow from 12% to 22% (WHO, 2022). It is estimated that the ageing population (aged over 65) in Northern Ireland (NI) will have exceeded 20.1% by 2028 (Age NI, 2021). Existing evidence has suggested that age is one of the strongest predictors of cognitive function (Deary et al., 2009); there are ~20,000 people with dementia in NI, and this is estimated to increase to 40,000 by 2040 (HSCNI, 2020). Thus, it is necessary to target the drivers of cognitive function in NI, especially for the ageing population.

Evidence is mounting that green space exposure is related to older adults' cognitive function (Cherrie et al., 2018; de Keijzer et al., 2016; Glover et al., 2024; Godina et al., 2023; Klompmaker et al., 2022; Hunter et al., 2023; Jimenez et al., 2022; Soloveva et al., 2023; Trott et al.,

This article is part of a special issue entitled: Aging in Space published in Health and Place.

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https://doi.org/10.1016/j.healthplace.2026.103660
Received 19 April 2025; Received in revised form 5 March 2026; Accepted 29 March 2026
Available online 15 April 2026
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2024). Existing studies found that green space exposure is associated with more physical activity and social interactions, both of which are linked to better cognitive health (Glover et al., 2024). An increasing number of epidemiological studies have examined the associations between green space exposure and cognitive function in high-income countries, but the results were inconsistent, partly due to the application of various green space measurements (de Keijzer et al., 2016). For example, Jimenez et al. (2022) used Normalised Difference Vegetation Index (NDVI), a satellite-based metric of the quantity of vegetation, to measure green space exposure and found it is positively associated with cognitive function in a US cohort. However, Chan et al. (2023) used the distance to the nearest public open space as the metric, but did not find evidence to support that it is associated with cognitive function in older adults in Hong Kong. Therefore, more empirical evidence using different measurements of green space exposure is still needed.

In recent years, it has become evident that there are great socioeconomic disparities in people's cognitive function, which means people with higher socioeconomic status (SES) usually have better cognitive function (Lyu and Burr, 2016; K. Wang et al., 2023). However, most of the existing studies mainly measured SES at elderhood rather than across the life course, so it is still unclear whether there are life-course socioeconomic disparities in cognitive function (Skoblow et al., 2024).

The 'equigenesis' theory indicates that people with lower SES or living in neighbourhoods with lower SES may be more affected by the availability of neighbourhood environments such as public amenities and green space, which means that green space may also have a stronger effect on the health of socioeconomic groups and thereby reducing SES inequalities in health (Mitchell and Popham, 2008; Mitchell et al., 2015; Nicholls et al., 2022). However, evidence of 'equigenesis' comes mainly from mental health or other physical health, such as obesity, while there is little evidence of cognitive function in older adults (Geary et al., 2023; Ngan et al., 2024; Rigolon et al., 2021). One study in the US found that there is no evidence that the effect of green space exposure on cognitive function in young adults varies among the different quartiles of Census tract median income (Jimenez et al., 2022).

Therefore, this study aims to explore whether green space exposure has stronger associations with cognitive function in later life for socioeconomically disadvantaged groups ('equigenesis') in a population of older adults in Northern Ireland. It particularly focuses on the extent to which life-course SES disparities moderate the association between green space and cognitive function in later life. We proposed two hypotheses: (1) more green space exposure is associated with better cognitive function in later life, and (2) the association between green space and cognitive function will be stronger among socioeconomically disadvantaged groups.

2. Data and methods

2.1. Study design and population

We used the first (between 2014 and 2016) and second (between 2017 and 2020) waves of The Northern Ireland Cohort for the Longitudinal Study of Ageing (NICOLA) conducted in NI (Neville et al., 2023). The NICOLA participants were selected based on multistage selection and geographic clustering of addresses. The Business Service Organisation General Practitioner Register Database was used to select participants' addresses, and then such information was merged with the Land and Property Services Pointer Database. Overall, there were 8478 and 6097 participants aged over 50 years included in the first and second waves of NICOLA, respectively. However, since the clinical cognition function test is labour-intensive, only a subgroup of participants took part in the clinical cognition function test (wave 1 = 3741, wave 2 = 5928). After data cleaning on key variables, we had a viable sample size of 3242 older adults in the first wave and 4971 in the second wave. More details of the study design can be found in Neville et al. (2023).

2.2. Cognitive function

Cognitive function is defined as multiple mental abilities such as learning and thinking (Tombaugh and McIntyre, 1992). Cognitive function was assessed based on the Mini-Mental State Examination (MMSE) scores (Tombaugh and McIntyre, 1992). MMSE consists of five dimensions of cognitive function tests (orientation, registration, attention and calculation, recall and language). The final scores range from 0 to 30, and higher scores suggest better cognitive function. More details of MMSE in NICOLA can be found in Frank Kee et al. (2021).

2.3. Life-course socioeconomic indicators

2.3.1. Early-life SES

We measured childhood SES using two indicators (self-reported childhood SES and unemployment of parents). First, participants were asked, 'Would you say your family during that time (before age 14) was pretty well off financially, about average, or poor?'. The responses were then classified into two groups (1 = 'poor childhood SES' vs. 0 = 'average/well childhood SES'). Since the self-reported wealth may not accurately reflect one's actual wealth, and this is especially the case for those above the average wealth ('deflated' status) (Kim and Sommet, 2025), we recoded and binarised the self-reported childhood SES variable to only compare the least wealthy to the average/above average. Also, the number of participants reporting 'pretty well off financially' is quite low compared to the other two answers, so keeping it as a stratified group may result in a relatively low level of estimation power. Second, participants were also asked, 'While you were growing up, before age 14, did your father/mother ever work outside the home?'. The responses were then classified into two groups (1 = 'both parents did not ever work outside the home' vs. 0 = 'others') since the single-earner family was common during the 1960s and 1970s in the UK (Smith, 2005).

2.3.2. Later-life SES

We measured adulthood SES using participants' educational attainment. Participants were asked 'What is the highest level of education you have completed?', and then we aggregated the responses into 'primary/none', 'secondary' and 'higher'.

Also, both the area-level Multiple Deprivation Measure and the capital value of the property for the current housing address were included. First, the Area Level Multiple Deprivation Measure was calculated using the Northern Ireland Index of Multiple Deprivation (NIMDM), which is based on different SES domains (NISRA, 2010). This index was created at the Super-Output Area level, which is the secondary level of administrative units in NI (average population = 790 persons). All Super-Output Areas were ranked from most deprived to least deprived and then split into five equal quintiles following previous studies (Neville et al., 2023). Then their quintile numbers were assigned to the Area level Multiple Deprivation Measure, and a higher quintile indicates a less deprived area (e.g., '1' suggests that participants lived in the most deprived area). Second, the capital value of the property for the current housing address was provided by Land and Property Services, which determined the level of local tax payable by each household (<£99,999, £100,000-£199,999, £200,000-£299,999 and >£300,000).

2.4. Green space exposure

2.4.1. Normalised Difference Vegetation Index (NDVI)

We used NDVI as a measure of the availability of vegetation (De La Iglesia Martinez and Labib, 2023). We used Sentinel-2 remote sensing images provided by the European Space Agency to calculate NDVI (Tucker, 1979) at 10 m × 10 m spatial resolution. The data were collected in the summer period (the greenest and least cloudy season) of 2015 and 2018 and then linked to the first and second waves of the NICOLA, respectively. We used the mean pixel values of NDVI within the 800-m circular buffer around the centroid of each housing address. We

have manually checked pixels with extremely high values (>0.9) and removed those that are not vegetation based on the Land Cover Map (Marston et al., 2023). NDVI values vary between −1 and 1, and a higher value indicates a higher density of healthy vegetation. The 800-m buffer was used for two reasons: 1) 10 to 15-min walking distance for the range of daily activities in the general population is usually assessed based on a 1000-m circular buffer, and 2) older adults tend to have a more constrained activity space than younger adults (Schwanen and Páez, 2010). We also used other buffer sizes to conduct several sensitivity analyses, and the results remained consistent across different buffer sizes (Table S1).

2.4.2. Accessibility of green space

We defined accessibility of green space as the distance (in meters) between participants' housing location and publicly accessible points of the nearest green space (e.g., parks and woodland). We included the accessibility of green space since it is a different green space metric compared with the availability of green space (NDVI). For example, existing studies found that accessibility of green space is more related to usage of the green space than the availability of green space (Young et al., 2012). Information on green space amenities was collected from the GreenspaceNI Map (https://www.out-scape.com/news/greenspaceni-map/) created by Outdoor Recreation Northern Ireland. The green space in the green space NI Map mainly refers to public and accessible parks and woodland. The distance to the nearest accessible points of parks and the distance to the nearest accessible points of woodland were calculated, respectively, for each of the participants in meters.

2.5. Covariates

Both waves of the NICOLA include information on demographic, socio-economic, and other individual data from participants. Following previous studies (Cherrie et al., 2018; de Keijzer et al., 2016; Jimenez et al., 2022; Ricciardi et al., 2022), we controlled for sex, age, marital status, employment, region (NISRA, 2015), smoking and drinking behaviour, the presence of cardiovascular diseases (CVDs), including hypertension, and duration of residency. Age was categorised into four groups (<60, 60-69, 70-79, and≥80) to enhance interpretability, and there is a nonlinear relationship between age and cognitive function (Deary et al., 2009). Also, the duration of residency was categorised into two groups (<5 years and≥5 years) since a 5-year exposure window is normally used to define long-term exposure for green space (Hystad et al., 2019). Summary statistics of variables are presented in Table 1.

2.6. Statistical analysis

To identify the association between green space exposure, life-course socioeconomic indicators and cognitive function in later life, we used multilevel linear regression models (Tom et al., 1999). Since MMSE scores in different waves were clustered within the same individuals and individuals were clustered in different Electoral Wards (subdivisions of the city) across the NI, we used a two-level random intercept model to allow the variances in intercepts across individuals and Electoral Wards. The test of Variance inflation factors (VIF<3) indicated that there is no multicollinearity among the independent variables.

As for the multilevel model, we first estimated the effect of the association between green space metrics, life-course socioeconomic indicators and MMSE scores in later life for the full sample using a multilevel linear regression model. Second, to test whether the association between green space metrics and MMSE scores vary across different SES groups, we stratified and reran the analysis by various life-course socioeconomic indicators (early-life SES and later-life SES). Last, we added interaction terms between green space metrics and life-course socioeconomic indicators to test whether the variances in the association between green space metrics and MMSE scores vary across different SES groups and are significant or not. In all contrasts, a p-value <0.05 was considered significantly different. We mainly focused on the significance level of coefficients for interaction terms. If the interaction terms are significant, then it suggests that the green space - MMSE scores association is different between the compared SES group and the referenced SES group. STATA v.15.1 was used for the statistical analysis (STATA, Inc., College Station, TX, USA). All models were conducted with a complete case analysis.

Table 1
Summary statistics of variables at the baseline.

Variable		Number	Value	%/SD
Sex	Female	1691		52.16
	Male	1551		47.84
Age	<60	1168		36.03
	60–69	1211		37.35
	70–79	672		20.73
	≥80	191		5.89
Marital status	Married/cohabiting/ same-sex civil partnership	2361		72.83
	Never married	209		6.45
	Separated/divorced/ widowed	672		20.73
Education	Primary/none	427		13.17
	Secondary	1459		45.00
	Higher	1356		41.83
Employment	Retired	1662		51.26
	Employed	1284		39.61
	Unemployed	71		2.19
	Others	225		6.94
Area-level Multiple Deprivation Measure	1 Most deprived	436		13.45
	2	583		17.98
	3	655		20.20
	4	671		20.70
	5 Least deprived	897		27.67
The capital value of the housing	<£99,999	892		27.51
	£100,000-£199,999	1667		51.42
	£200,000-£299,999	504		15.55
	>£300,000	179		5.52
Childhood socioeconomic status	Poor	619		19.09
	Average or well	2623		80.91
Childhood unemployment of parents	Yes	60		1.85
	No	3182		98.15
Region	Urban	583		17.98
	Intermediate	1822		56.20
	Rural	837		25.82
The presence of CVDs	Yes	1696		52.31
	No	1546		47.69
Current smoker	Yes	546		16.84
	No	2696		83.16
Current drinker	Yes	2239		69.06
	No	1003		30.94
Duration of residency	<5 years	163		5.03
	≥ 5 years	3079		94.97
MMSE Scores			28.47	1.84
Green space	NDVI		0.36	0.17
	Distance to the nearest parks (m)		3139.98	4031.64
	Distance to the nearest woodland (m)		3336.20	2400.17

Note. SD = standard deviation; NDVI = normalised difference vegetation index; MMSE = mini-mental state examination.

3. Results

3.1. Descriptive characteristics of the baseline (Wave 1) study population

Table 1 summarises the characteristics of the participants in the first wave. The mean MMSE scores for the baseline population were 28.47 (SD = 1.84), while the average NDVI, distance to the nearest parks and distance to the nearest woodland were 0.36 (SD = 0.17), 3139.98 m

(SD = 4031.64) and 3336.20 m (SD = 2400.17), respectively. About one-fifth of the participants (19.09%) reported poor childhood SES, while only 1.85% had childhood unemployment of parents. Nearly 13.17% of the participants had a primary/none qualification, 45.00% had a secondary qualification, and 41.83% had a higher qualification. Approximately 13.45% lived in the most deprived areas, and 27.67% lived in the least deprived areas. Also, more than half of the participants' capital value of the housing was below £200,000 (78.93%).

Overall, 52.16% of the participants were female, and the mean age was 63.95 (SD = 8.93). 20.73% of the participants were separated/divorced/widowed, and approximately half of the participants (51.26%) were retired. 17.98% lived in urban regions, and most of the participants (94.97%) had a duration of residency of over five years. Furthermore, just over half of the participants (52.31%) had CVDs, 16.84% were current smokers, and 69.06% were current drinkers.

3.2. The result of the multilevel models in the associations between green space metrics and MMSE scores

Table 2 shows the result of the multilevel model. The value of the Akaike information criterion is larger in the unadjusted model than in the adjusted model, so controlling for covariates improves the accuracy of the model. The result suggests that NDVI was positively linked to MMSE scores (Coef = 0.51, SE = 0.13), and distance to the nearest parks was negatively associated with MMSE scores (Coef = −0.09, SE = 0.02). This means every unit increase in NDVI is associated with a 0.51 unit increase in MMSE scores, while every unit increase in distance to the nearest parks is associated with a 0.05 unit decrease in MMSE scores in older adults in NI. However, there is no evidence that distance to the nearest woodland (Coef = 0.01, SE = 0.03) was associated with MMSE scores. The result for decomposition of variance shows that the variance between individuals was the highest (1.88), while the variance between Wards was the lowest (0.25), which means that variance between individuals accounted for more total variation in participants' MMSE scores than variance between Wards and variance within individuals (i.e., the variance between the first and second wave for the same individual).

3.3. Green space metrics, life-course socioeconomic indicators and MMSE scores

Fig. 1 and Table S2 show the association between green space metrics and MMSE scores stratified by different SES groups. First, Fig. 1a indicates that NDVI was positively associated with MMSE scores for both participants reporting poor childhood SES (Coef = 0.559, SE = 0.225) and those reporting average or well childhood SES (Coef = 0.495, SE = 0.144). Also, distance to the nearest parks was negatively associated with MMSE scores for both participants reporting poor childhood SES (Coef = −0.101, SE = 0.049) and those reporting average or well childhood SES (Coef = −0.079, SE = 0.024). However, the results in Table 3 suggest that there is no evidence that the association between NDVI/distance to the nearest parks and MMSE scores was significantly different between participants reporting poor childhood SES and those reporting average or well childhood SES.

Second, Fig. 1b indicates that NDVI was positively associated with MMSE scores for both participants reporting childhood unemployment of parents (Coef = 1.105, SE = 0.210) and those not reporting childhood unemployment of parents (Coef = 0.501, SE = 0.132). Also, distance to the nearest parks was negatively associated with MMSE scores only for those not reporting childhood unemployment of parents (Coef = −0.085, SE = 0.022). The results in Table 3 further suggest that the association between NDVI and MMSE scores was significantly stronger for participants reporting childhood unemployment of parents than those not reporting childhood unemployment of parents.

Third, Fig. 1c indicates that NDVI was positively associated with MMSE scores for participants with primary/no qualification (Coef = 0.828, SE = 0.377), participants with secondary qualification (Coef = 0.560, SE = 0.199), and those with higher qualification (Coef = 0.319, SE = 0.154). Also, distance to the nearest parks was negatively associated with MMSE scores for both participants with primary/none qualification (Coef = −0.140, SE = 0.070) and participants with secondary qualification (Coef = −0.089, SE = 0.031). The results in Table 3 further suggest that the association between distance to the nearest parks and MMSE scores was significantly stronger for participants with primary/none qualifications than those with higher qualifications.

Fourth, Fig. 1d indicates that NDVI was positively associated with MMSE scores for both participants with a capital value of housing below £99,999 (Coef = 0.833, SE = 0.298) and participants with a capital value of housing between £100,000 and £199,999 (Coef = 0.410, SE = 0.172). Also, distance to the nearest parks was negatively associated with MMSE scores for both participants with a capital value of housing below £99,999 (Coef = −0.094, SE = 0.042) and participants with a capital

Table 2
Regressing green space exposure and socioeconomic indicators on MMSE scores in NI.

	Unadjusted: Coef. (SE)	Adjusted: Coef. (SE)
NDVI	**0.23**(0.12)**	**0.51***(0.13)**
Distance to the nearest parks	**−0.05**(0.02)**	**−0.09***(0.02)**
Distance to the nearest woodland	0.03(0.04)	0.01(0.03)
Sex (ref = Female)		
Male		**−0.31***(0.05)**
Age (ref = < 60)		
60–69		−0.10*(0.06)
70–79		**−0.35***(0.07)**
>80		**−1.15***(0.10)**
Marital status (ref = Married/cohabiting/same-sex civil partnership)		
Never married		**−0.27***(0.09)**
Separated/divorced/widowed		**−0.29***(0.06)**
Education (ref = Primary/none)		
Secondary		**0.93***(0.07)**
Higher		**1.37***(0.07)**
Employment (ref = Retired)		
Employed		**0.27***(0.06)**
Unemployed		−0.05(0.15)
Others		**−0.49***(0.09)**
Area level Multiple Deprivation Measure (ref = 1 Most deprived)		
2		0.11(0.08)
3		0.08(0.08)
4		**0.22***(0.08)**
5 Least deprived		**0.23***(0.08)**
Capital value of the housing (ref = <£99,999)		
£100,000-£199,999		**0.22***(0.06)**
£200,000-£299,999		**0.27***(0.08)**
>£300,000		0.21*(0.12)
Childhood socioeconomic status (ref = Average or well)		
Poor		−0.06(0.06)
Childhood unemployment of parents (ref = No)		
Yes		−0.01(0.16)
Duration of residency (ref = < 5 years)		
≥ 5 years		0.14(0.10)
Region (ref = Urban)		
Intermediate		−0.03(0.07)
Rural		−0.09(0.08)
Current drinker (ref = current non-drinker)		**0.32***(0.05)**
Current smoker (ref = current non-smoker)		0.04(0.06)
With CVDs (ref = without CVDs)		0.05(0.13)
Variance (Ward level)	0.12	0.25
Variance (Between individuals)	2.28	1.88
Variance (Within individuals)	1.77	1.37
AIC	33903.69	32574.81

Note: NDVI = normalised difference vegetation index; Coef = coefficient; SE = standard error; AIC = Akaike information criterion. *p < 0.10, **p < 0.05, ***p < 0.01.

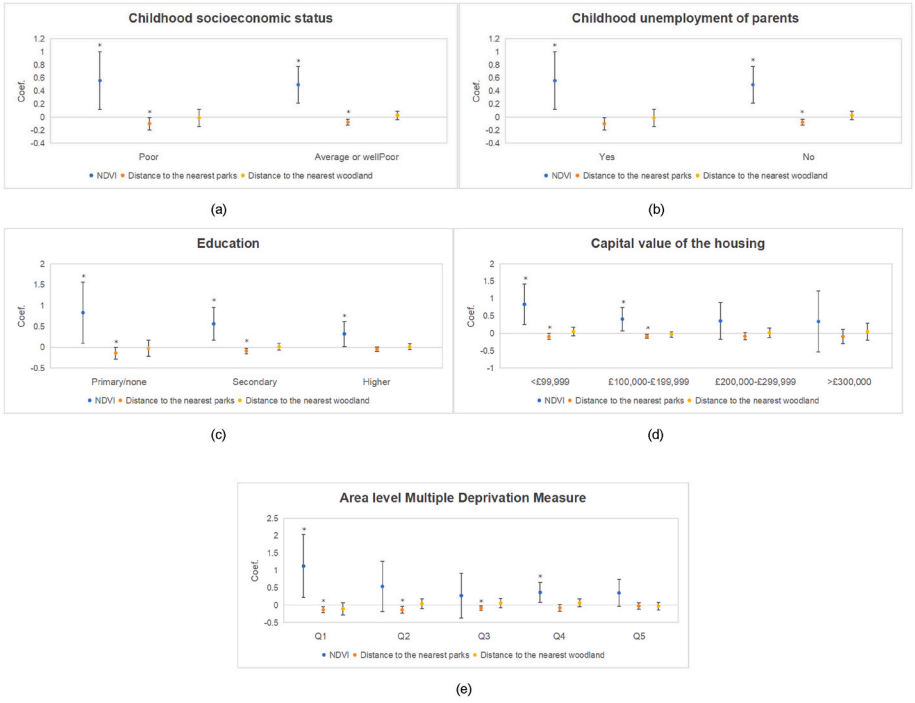


Fig. 1. Life-course SES disparities in the association between green space metrics and MMSE scores in NI. Models adjusted for covariates. NDVI = Normalised Difference Vegetation Index. *p < 0.05.

Table 3
Tests for differences in the association between green space metrics and MMSE scores among different SES groups.

		P for interaction		
		NDVI	Distance to the nearest park	Distance to the nearest woodland
Childhood socioeconomic status (ref: Average or well)	Poor	0.596	0.451	0.472
Childhood unemployment of parents (ref: No)	Yes	0.047**	0.830	0.213
Education (ref: Primary/none)	Secondary Higher	0.272	0.011**	0.611
Area level Multiple Deprivation Measure (ref: 1 Most deprived)	2 3 4 5 Least deprived	0.010**	0.014**	0.283
Capital value of the housing (ref: <£99,999)	£100,000-£199,999 £200,000-£299,999 >£300,000	0.046**	0.897	0.569

Note: Models adjusted for covariates. NDVI = normalised difference vegetation index; Coef = coefficient; SE = standard error. *p < 0.10, **p < 0.05, ***p < 0.01.

value of housing between £100,000 and £199,999 (Coef = −0.090, SE = 0.029). The results in Table 3 further suggest that the association between NDVI and MMSE scores was significantly stronger for participants with a capital value of housing below £99,999 than for those with a capital value of housing above £99,999.

Last, Fig. 1e indicates that NDVI was positively associated with MMSE scores for participants living in areas with the 1st and 4th quintiles of Area level Multiple Deprivation Measure (1st: Coef = 1.128, SE = 0.462; 4th: Coef = 0.370, SE = 0.148). In contrast, distance to the nearest parks was negatively associated with MMSE scores for participants living in areas with the 1st, 2nd, and 3rd quintiles of Area level Multiple Deprivation Measure (1st: Coef = −0.134, SE = 0.043; 2nd: Coef = −0.133, SE = 0.048; 3rd: Coef = −0.081, SE = 0.034). The results in Table 3 further suggest that the association between NDVI/distance to the nearest parks and MMSE scores was significantly stronger for participants living in areas with the 1st quintile of Area level Multiple Deprivation Measure than those living in areas with higher quintiles of Area level Multiple Deprivation Measure.

4. Discussion

This study extends previous research on the association among green space, life-course SES disparities and cognitive function ('equigenesis'). The most important finding in this study is that the effect of green space is stronger in socioeconomically disadvantaged groups and thus may reduce life-course SES disparities in cognitive function in later life (potential 'equigenesis' in cognitive function).

4.1. Main findings

We found that higher green space exposure (e.g., higher NDVI or living closer to parks) at older adults' housing locations was positively associated with their cognitive function, which is consistent with existing evidence (Cherrie et al., 2018; Jimenez et al., 2022; Ricciardi et al., 2022). Most importantly, the result further indicated that green space exposure has a stronger effect on socioeconomically disadvantaged groups, which means it may contribute to fewer life-course socioeconomic disparities in cognitive function for older adults.

First, as for early-life SES, green space exposure is more effective for older adults with lower childhood SES. One possible explanation is that socioeconomically disadvantaged children usually have fewer health-related resources, such as lower availability of green space in different contexts (McEachan et al., 2018). Previous studies have found that there is a dose-response association between green space exposure and health outcomes (Mygind et al., 2021), so older adults with higher childhood SES may enjoy more childhood green space exposure and accumulated

green space exposure through the life course, which makes them less sensitive to the green space exposure during the elderhood. Another possible explanation is that people with lower childhood SES are more likely to experience stressful life events such as unexpected financial distress of parents, and these stressors may also be accumulated through the life course, which may finally influence health during elderhood (Zimmer et al., 2016). Existing evidence found that green space can buffer the association between stressful life events and health since visiting green space can help people restore energy and relieve stress (Van den Berg et al., 2010). Therefore, green space exposure during elderhood can be more effective for older adults with lower childhood SES.

Second, as for later-life SES, green space exposure has a stronger influence on older adults with lower educational attainment. This may be because older adults with lower educational attainment may be less physically active due to a lack of health-related knowledge, while green space may encourage more physical activity and reduce the inequalities in health caused by educational attainment (Hunter et al., 2019; Rigolon et al., 2021). Also, the insufficiency of health-related knowledge may make older adults with lower educational attainment less aware of the benefits of viewing and visiting green spaces, which may further lead to inequalities in health (Wang et al., 2024; Wang et al., 2022; Wang et al., 2025). However, with more green space available within the neighbourhood, older adults with lower educational attainment may enjoy more unconscious green space exposure, even if lacking in health-related knowledge, which may reduce the potential inequalities (Wang et al., 2022, 2024).

Also, the association between green space exposure and cognitive function is more pronounced for older adults with a lower capital value of housing or living in more deprived neighbourhoods. Older adults in deprived neighbourhoods usually have fewer medical resources (e.g., hospitals) to support their health, so health-supportive amenities such as green space can be more important in such areas (Rigolon et al., 2021). In addition, since vegetation such as street trees may be able to block air pollutants and traffic noises (Klingberg et al., 2017), it can be more important for older adults' health (e.g., cognitive function) in more deprived neighbourhoods where the level of pollution such as air pollution and noise is higher (Tonne et al., 2018; Young et al., 2012).

Our findings have important implications for promoting healthy ageing and reducing health inequalities. First, the results show that the higher level of NDVI and better access to parks are associated with better cognitive function in older adults. This suggests that policymakers should try to improve the provision of vegetation and older adults' access to public parks. For example, the NI government has proposed a target that 90% of the population should have access to public green space within 400m (Northern Ireland Statistics and Research Agency, 2023). Second, our findings suggest that socioeconomically disadvantaged older adults may benefit more from green space exposure for their cognitive function. Therefore, more attention should be paid to socioeconomically disadvantaged older adults. More investment in green space amenities should be implemented in more deprived areas, which also have a large number of the ageing population. Also, since both early-life and later-life SES disparities exist, it is necessary to pay attention to SES disparities across different cohorts when the disparities have not accumulated yet.

4.2. Limitations

The following limitations should be noted. First, there are only two waves of NICOLA data available, but this is not enough to reflect variances in cognitive function over time. Second, although MMSE scores have been validated in the NI population (Neville et al., 2023), more metrics of cognitive function in older adults should be used to cross-validate the findings. Also, MMSE scores are influenced by people's education level, which results in people with higher educational attainment scoring higher without actually having better cognitive function (Brucki et al., 2011). Since we used education as one of the SES indicators, our findings on the educational disparities in the green space-cognitive function may simply reflect that such an association may vary by the levels of cognitive function rather than SES. For example, there is greater variation in MMSE scores of older adults with lower levels of education than those with higher levels of education in our sample (Primary/none: SD in MMSE = 2.42, range in MMSE = 15-30; Secondary: SD in MMSE = 1.79, range in MMSE = 16-30; Higher: SD in MMSE = 1.29, range in MMSE = 18-30; p value in variance-comparison test <0.001). Therefore, while controlling education in analyses does help address some of the effects of educational bias in MMSE scores (Jones and Gallo, 2001), it does not fully address ceiling and floor effects. Future research should consider using alternative measures of cognitive function. Third, existing studies have documented that green space quality also matters for health (Wang et al., 2022). Still, we only included metrics of green space quantity, which prevents us from having a systematic understanding of the health benefits of green space exposure. Also, we did not measure green space exposure in childhood and adulthood, which prevents us from understanding the effect of green space exposure from a life-course perspective. The NDVI metric may be sensitive to seasonal changes, so assessing NDVI in the greenest season may overestimate its health benefits (De La Iglesia Martinez and Labib, 2023). Fourth, some of the metrics of socioeconomic disparities are self-reported, which may lead to bias. Also, we only have a limited number of metrics of socioeconomic disparities, while some important metrics, such as parental educational attainment, are missing due to data availability. Hence, we did not use a consistent socioeconomic indicator across different life stages, which makes socioeconomic indicators less comparable in this study. Fifth, the selection of covariates may also lead to bias. For example, some covariates may also have a mediating role in the association between green space and cognitive function, so our model can be over-adjusted. Last, we were not able to fully address residential self-selection bias (Boone-Heinonen et al., 2010). For example, older adults who had some unobserved attributes (e.g. nature enthusiasts) were in favour of greener neighbourhoods and were more likely to have better cognitive function.

5. Conclusion

This is one of the first studies to explore whether socioeconomically disadvantaged groups may benefit more from green space exposure for their cognitive function ('equigenesis') from a life-course perspective. Results show that higher green space exposure may help to reduce life-course socioeconomic disparities in cognitive function in later life, since green space-cognitive function associations were stronger in socioeconomically disadvantaged groups. To achieve the goal of better cognitive function in later life and reducing socioeconomic disparities in cognitive function through urban planning and design, policymakers and planners should consider enhancing the provision of green space, especially in disadvantaged neighbourhoods.

Ethical approval

Written informed consent was obtained from participants prior to participation under ethical approval from the School of Medicine, Dentistry and Biomedical Sciences Ethics Committee, Queen's University Belfast (SREC 12/23) and in accordance with the Helsinki Declaration.

Funding

This work is part of the SPACE (Supportive Environments for Physical and Social Activity, Healthy Ageing and Cognitive Health) research project, which was supported by the Economic and Social Research Council and Innovate UK under the Healthy Ageing Challenge Social, Behavioural and Design Research Programme fund [ES/V016075/1].

This work was also supported by the UK Prevention Research

Partnership (MR/V049704/1), which is funded by the British Heart Foundation, Cancer Research UK, Chief Scientist Office of the Scottish Government Health and Social Care Directorates, Engineering and Physical Sciences Research Council, Economic and Social Research Council, Health and Social Care Research and Development Division (Welsh Government), Medical Research Council, National Institute for Health Research, Natural Environment Research Council, Public Health Agency (Northern Ireland), The Health Foundation and Wellcome. The work is also supported by the HSC Research and Development Office Northern Ireland (COM/5634/20), and National Institutes of Aging (R01AG030153).

The Atlantic Philanthropies, the Economic and Social Research Council, the UKCRC Centre of Excellence for Public Health Northern Ireland, the Centre for Ageing Research and Development in Ireland, the Office of the First Minister and Deputy First Minister, the Health and Social Care Research and Development Division of the Public Health Agency, the Wellcome Trust/Wolfson Foundation and Queen's University Belfast provided core financial support for NICOLA Waves 1 and 2. The authors alone are responsible for the interpretation of the data and any views or opinions presented are solely those of the authors and do not necessarily represent those of the NICOLA Study team.

CRediT authorship contribution statement

Ruoyu Wang: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Hüseyin Küçükali:** Writing – review & editing. **Ione Avila-Palencia:** Writing – review & editing. **Shay Mullineaux:** Writing – review & editing. **Jennifer McKinley:** Writing – review & editing. **Frank Kee:** Writing – review & editing. **Bernadette McGuinness:** Writing – review & editing. **Ruth F. Hunter:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Conflicts of interests

The authors declare no competing interests.

Acknowledgements

We are grateful to all the participants of the NICOLA Study, and the whole NICOLA team, which includes nursing staff, research scientists, clerical staff, computer and laboratory technicians, managers and receptionists. We also want to thank Angie Scott, Siva Padmanabhan, Maciej Domanski and Charlotte Neville for their support for data management.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.healthplace.2026.103660.

Data availability

Researchers can apply for access to the data by submitting a Research proposal to the NICOLA Data Access Committee. For more information, please refer to https://www.qub.ac.uk/sites/NICOLA/InformationforResearchers/.

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